

Enhancing Tumor Segmentation with Conformal Prediction

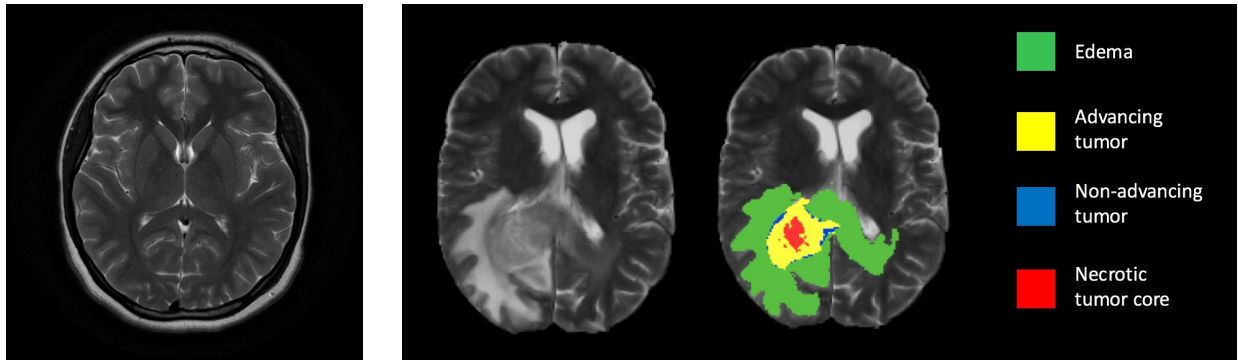
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Motivation

- Simple Accuracy in prediction is not enough!

- Rule in
or
- rule out



- harmful diagnoses

$$\mathbb{P}[\text{true diagnosis} \in \{\textit{Normal}, \textit{Concussion}, \textit{Cancer}\}] \geq 90\%$$

Conformal Prediction Theorem

Create valid prediction interval, $C(X_{n+1})$, such that:

$$1 - \alpha \leq [Y_{n+1} \in C(X_{n+1})] \leq 1 - \alpha + \frac{1}{n+1}$$

For any **Algorithm**, **Dataset**, α , n (calibration).

Conformal Risk Control

We create valid prediction sets $C_\lambda(X_{n+1})$ by choosing the *set* scale parameter λ such that the expected false negative rate α is controlled.

Controlled conformal risk:

$$\mathbb{E}[\ell(C_\lambda(X_{n+1}), Y_{n+1})] \leq \alpha$$

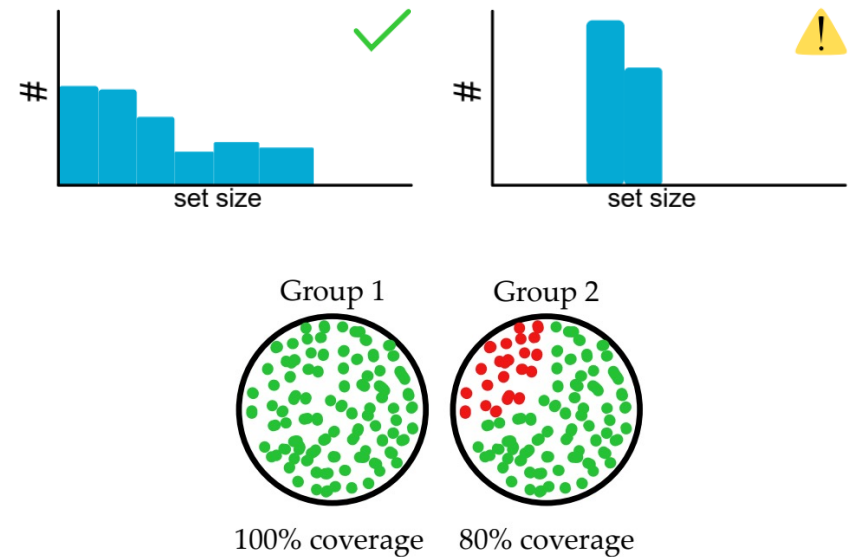
Risk metric:

$$L_i^{FNR}(\lambda) = 1 - \frac{|Y_i \cap C_\lambda(X_i)|}{|Y_i|}$$

For any **Algorithm**, **Dataset**, α , n (calibration).

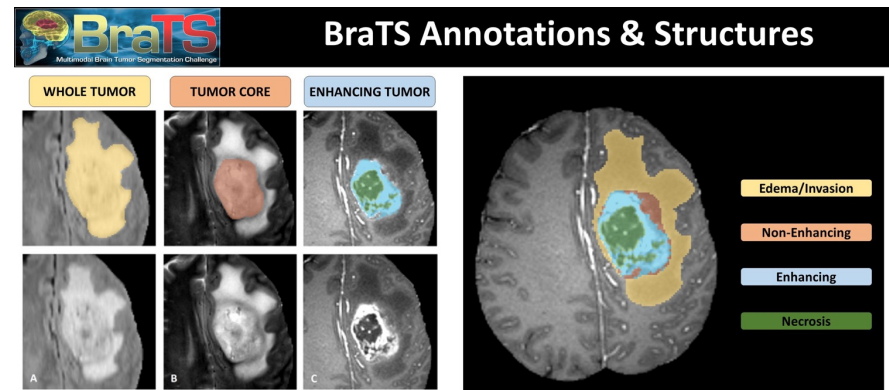
Evaluating Conformal Prediction

- Set size
 - Good Conformal Prediction is *adaptive* to the model's uncertainty.
- Correctness check
 - Correctness checks help test if model is correctly implemented

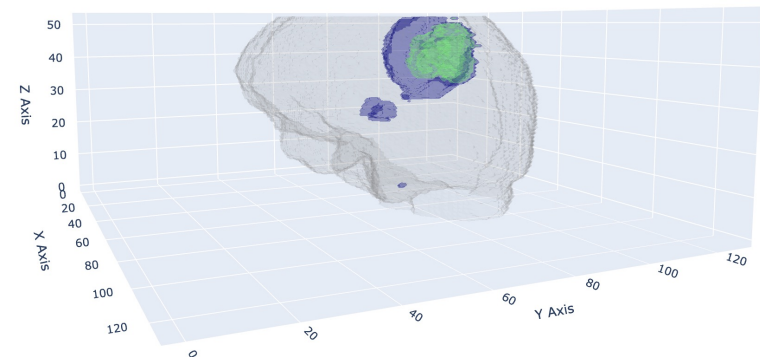


Methodology

- Base model: top ranked tumor segmentation model on the BraTS2020 dataset on kaggle by [Rastislav Kopál](#)
- calibrate Adaptive Conformal Prediction model outlined by Romano, Y., Sesia, M., & Candès, E. J. (2020).
- Extending it with FNR control Angelopoulos, A. N., Bates, S., Fisch, A., Lei, L., & Schuster, T. (2022).



Result:



Result

- HTML:

Model Validation Performance

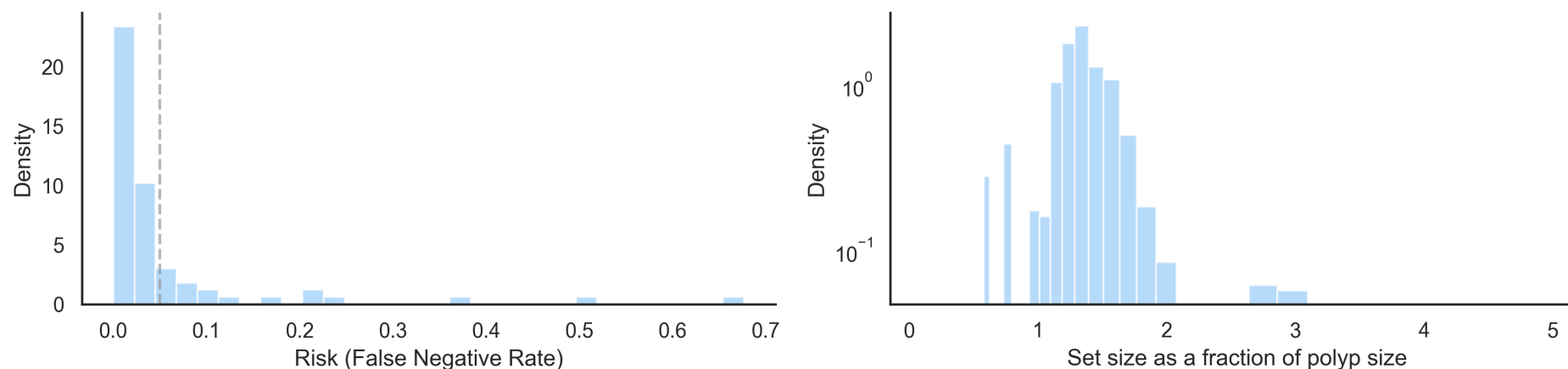
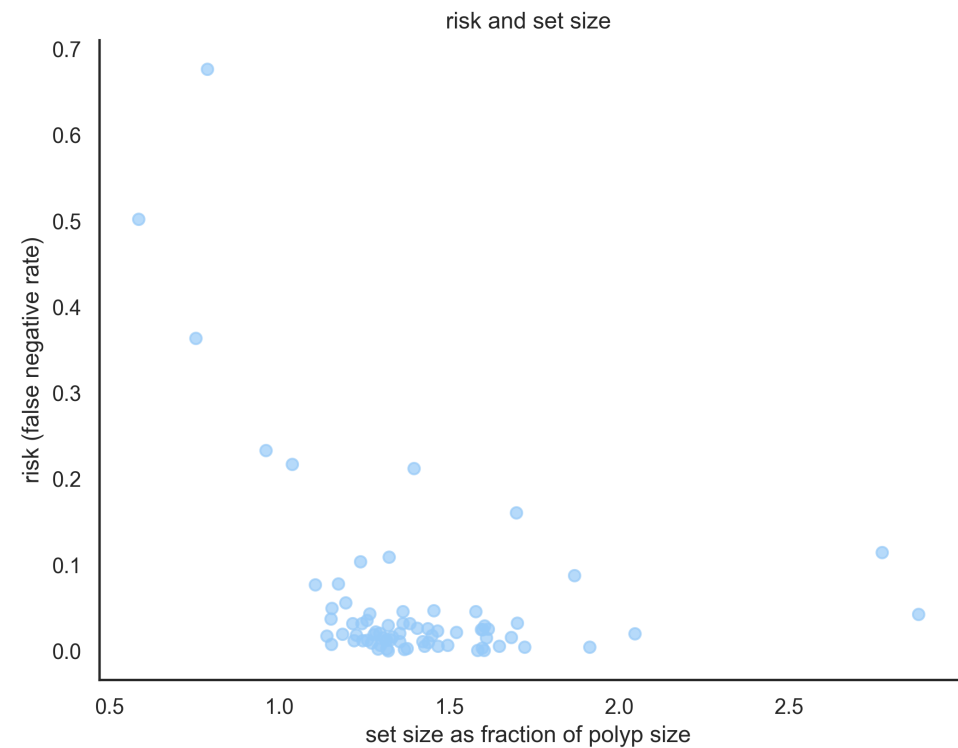


Figure 1: **FNR control in tumor segmentation.** The plot reports the FNR and set size over 74 independent random data splits. The gray dashed line marks the target false negative rate $\alpha = 0.05$. The mean and standard deviation of the risk over the validation set are 0.0558 and 0.1095 respectively.

Model Validation Performance

Figure 2: **FNR control in tumor segmentation.** The plot reports the relationship between the FNR and set size over 74 independent random validation samples.



Relevance

Doctor can control the conformal predictor.



Conformal Predictor can control the Doctor.

- Doctor checks high uncertainty predictions by hand.
- Saving valuable time, allowing for more patients to be treated by one doctor, while controlling risk

- Conformal Predictor can detect false negative cases from patients marked 'healthy' by doctor. Saving lives!

References

Angelopoulos, A. N., Bates, S., Fisch, A., Lei, L., & Schuster, T. (2022). Conformal risk control. arXiv:2208.02814. <https://doi.org/10.48550/arXiv.2208.02814>

Angelopoulos, A. N., & Bates, S. (2021). A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. arXiv preprint arXiv:2107.07511. Retrieved from <https://arxiv.org/abs/2107.07511>

Menze, B. H., et al. (2015). The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS). *IEEE Transactions on Medical Imaging*, 34(10), 1993-2024. <https://doi.org/10.1109/TMI.2014.2377694>. PMID: 25494501; PMCID: PMC4833122.

Romano, Y., Sesia, M., & Candès, E. J. (2020). Classification with valid and adaptive coverage. arXiv:2006.02544. Retrieved from <https://arxiv.org/abs/2006.02544>